- *Alerting/visualization*—Collecting data about the state of a system is useful only if it's used to take some action. When a failure is identified while monitoring an application, an alert should be triggered, and some action should be taken to handle it. Specific dashboards are used to visualize the data collected and plot them in relevant graphs to provide a good picture of the system's behavior. This book will show how to use Grafana to visualize data collected from cloud native applications.
- *Distributed systems tracing infrastructure*—In a distributed system, it's not enough to track every single subsystem's behavior. It's essential to trace the data flowing through the different subsystems. In this book, you'll integrate Spring with OpenTelemetry and use Grafana Tempo to collect and visualize the traces.
- *Log aggregation/analytics*—Keeping track of the main events in an application is critical to infer the software's behavior and debug it if something goes wrong. In a cloud native system, logs should be aggregated and collected to provide a better picture of the system's behavior and ensure the possibility of running analytics to mine information from that data. Throughout the book, I'll talk more about logs. You'll use Fluent Bit, Loki, and Grafana to collect and visualize logs, and you'll learn the best practices for logging in a cloud native context.

1.3.5 Manageability

In control theory, the counterpart of observability is controllability—the ability of external input to change the state or the output of a system in a finite time interval. This concept leads us to the last of the main properties of cloud native: manageability.